Heat-engine efficiency

antocciphysics

UNIT 5 / DAY 9 / HONORS + SEI

Name: ______________________________ **Block:** ________

(teacher only)
Mastery Rating: ____________

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting
② Can do part; need help
③ Solve and explain with allowed supports

LANGUAGE GOAL Use “efficiency” and “cycle” to explain your model or result. Sentence starters are optional.

efficiency: useful output divided by input
cycle: a process that returns a system to its initial state
kelvin: the absolute temperature scale used in thermal ratios

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01 Bring back an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes 9 times as large.
Compare rms speed.

Word help: particle = a small constituent, such as an atom or molecule

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = 1 - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

Worked reasoning: A heat engine takes 100 J from a hot source, does 25 J of work and rejects 75 J. Efficiency=25/100=25%.

Your turn: Draw three labeled energy arrows and check conservation.

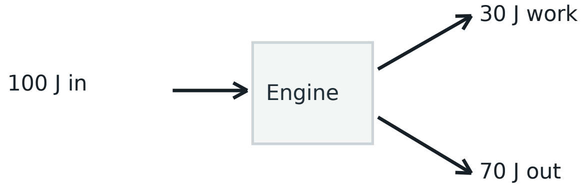
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

An engine receives 100 J, produces 30 J of work and rejects 70 J.
Check the energy balance and find efficiency.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Solve and justify

An engine operates between 600 K and 300 K and receives 800 J per cycle.
Find ideal maximum efficiency and maximum work.
Actual work is 240 J; find actual efficiency.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why should a 95% efficient electric motor not be confused with a 95% heat engine converting heat from one source?

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Choose a method

At fixed speed, mass becomes 4 times as large while energy-coupling fraction becomes 1/6 as large. **Compare** coupled energy.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

07 Exit check

A heat engine has Q_{hot}=1500 J and Q_{cold}=1050 J. **Find** work and efficiency.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

An ideal upper bound is $\eta=1-T_c/T_h$ with kelvin temperatures. For T_h=600 K and T_c=300 K, compare that bound with an engine producing 400 J work from 1000 J input.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.